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SECTION B

Answer **all** questions.

8. The Mohs scale measures the hardness of different materials and runs from 1 to 10, with 10 being the hardest. A fingernail is rated as 2.5 on the Mohs scale so any material that can be scratched with a fingernail has a hardness of less than 2.5. Graphite and the minerals halite and tachyhydrite, can all be scratched with a fingernail.

(a) The hardness value of graphite is approximately 1.5.

Describe the structure of graphite and explain why it is soft. [2]

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(b) Halite is the mineral name for rock salt. It contains sodium chloride.

The equation corresponding to the standard enthalpy change of formation of sodium chloride is as follows.

$$Na(s) \quad + \quad \tfrac{1}{2}Cl_2(g) \longrightarrow NaCl(s) \qquad \Delta_f H^{\theta} = -411\ \mathrm{kJ\ mol^{-1}}$$

Some standard enthalpy changes are given in the table.

Enthalpy term	Standard molar enthalpy change / $\mathrm{kJ\ mol^{-1}}$
first ionisation energy of sodium	494
electron affinity of chlorine	–364
bond energy of Cl_2	242
enthalpy of atomisation of sodium	109

(i) State the enthalpy of atomisation of chlorine. [1]

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(ii) Use the standard enthalpy changes to find the standard enthalpy change of lattice formation of sodium chloride. [3]

Standard enthalpy change of lattice formation = ... $kJ\,mol^{-1}$

(iii) A student suggests that the entropy change for the formation of sodium chloride must be positive as the reaction occurs easily.

Is the student correct? Justify your answer. [2]

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(c) Tachyhydrite is a soft, soluble mineral that has a general formula of $A_aB_bX_c.dH_2O$. A and B represent two different s-block metal ions, with X representing an anion.

Use **all** the following data to find the identity of all the ions present and to find the formula of tachyhydrite.

A, B and X are not the chemical symbols of the elements present.

A flame test on a sample of tachyhydrite gives a brick-red flame with no indication of any other colour. Addition of dilute sulfuric acid to a solution of tachyhydrite causes the solution to go cloudy. With concentrated sulfuric acid, misty fumes are released from the solid, however there is no colour change. Addition of aqueous silver nitrate to a solution of tachyhydrite causes a precipitate to form.

Tachyhydrite has an M_r of approximately 518. On heating to constant mass the solid loses 41.75 % of its initial mass. [6 QER]

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Formula of tachyhydrite ....

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